layers of ESM2 to output the logits of the mutated sequence, $\mathbf{x}_{\text{logits,mut}} \in \mathbb{R}^{L\times V}$, where L is the sequence length and V is the vocabulary size.

We compare these new logits to the logits of the wild-type sequence, $\mathbf{x}_{\text{logits,wt}} \in \mathbb{R}^{L\times V}$. For each amino acid position, we calculate the cosine similarity between the respective logit vectors. We only accept a mutation at a given position if the cosine similarity is below 0.98, ensuring that we only mutate amino acids where ESM2 has made a meaningful change.

To find the optimal multiplier, we perform a grid search over values from -3 to 3 with a step size of 0.2. For each multiplier, we use the linear probe to predict the fitness of the resulting sequence. We then select the top 50 unique sequences with the highest predicted fitness. If a sequence has been previously designed, we move to the next highest-scoring sequence to ensure we design 50 unique variants.

Simulated Annealing. We adapt the simulated annealing code from `https://github.com/gitter-lab/metl-pub/tree/main/sim-annealing` [27]. All parameters are left as default. We run simulated annealing over the linear probes trained on the ESM layer and ESM logits. The number of mutations per designed sequence was determined by sampling from the Poisson distribution $\text{Pois}(2) + 1$, ensuring that the maxmimum number of mutations possible is still five. To ensure a fair comparison, we ensured both feature steering and simulated annealing took a comparable amount of time. We set the number of simulated annealing timesteps based on the time required for feature steering to design 50 variants. This was done by first measuring the time needed to complete 1,000 simulated annealing timesteps and then scaling accordingly.

Random. To create the random baseline, we sample from the Poisson distribution of $\text{Pois}(2) + 1$ to determine the number of mutations to make. We then choose the mutated amino acid uniformly at random.

A.5 MLP Training

To create a fitness prediction model for our protein engineering tasks, we trained an MLP for each DMS assay. The MLP is a three-layer feedforward network with ReLU activation functions, taking in a flattened one-hot encoding of the entire protein sequence as input. The network architecture consists of an input layer, a hidden layer with 128 neurons, a second hidden layer with 64 neurons, and a final output layer with a single neuron to predict the fitness score.

For each DMS assay, we split the full data into a training set (80%) and a validation set (20%). We then trained the MLP for up to 1000 epochs using the AdamW optimizer with a learning rate of 10^{-3} and Mean Squared Error (MSE) as the loss function. To prevent overfitting, we employed early stopping with a patience of 10 epochs based on the validation loss.

A.6 Feature Visualization

Given the wildtype sequence, we find the latent representation $\mathbf{z} \in \mathbb{R}^{d_{\text{SAE}}\times L}$ by passing the sequence through layer 24 of ESM2 to get the embeddings and then passing the embeddings through the SAE encoder. We use the linear probe weights and find the indices that correspond to the five largest positive and negative probe weight indices for which the corresponding index in $\mathbf{z}$ is active as well. We then find the amino acids in the sequence that are being activated by the SAE: given the i^{th} latent, the amino acid activations associated with this latent are $\mathbf{z}[i,:]$. We then use the top five mutants with the highest fitness found from steering the SAE and analyze the activation difference to find the amino acids in the sequence that had the largest absolute activation difference between the wild-type and steered sequence SAE embeddings. We use PyMOL to visualize these changes.

To identify active sites in GFP, we utilize the positions provided in [30] under the Methods section titled "Refinement and mutational scan". For GB1, we identify allosteric and binding sites based on [25] from Extended Data Fig. 7c. We additionally identify epistatic sites based on [23].

A.7 Weight Sparsity

To quantify sparsity in linear probe weights, we measure the proportion of total variance explained by the top 5% of weights ranked by magnitude. Using linear probe weights from the random extrapolation task with the first seed, we compute, for each training size N, the ratio between the variance captured by the top 5% of weights and the total variance of all weights (where the total number of weights is d_{SAE} for SAE, d_{ESM} for ESM layer, and V for ESM logits) is computed. We plot the magnitude of probe weights for each model in Fig. 4. For visualization purposes, we exclude weights that have a magnitude greater than 3. This occurs 11 times in the ESM logits but not in the ESM layer or SAE.

B Additional Experimental Results

Table 4: Average Spearman ρ across all low-N regimes under mutation extrapolation.

Method	**DMS**	N **= 8** $\uparrow$	N **= 24** $\uparrow$	N **= 96** $\uparrow$	N **= 384** $\uparrow$
SAE	GFP_AEQVI_Sarkisyan	0.06 ± 0.09	$\mathbf{0.15 \pm 0.10}$	$\mathbf{0.29 \pm 0.08}$	$\mathbf{0.36 \pm 0.04}$
	SPG1_STRSG_Olson	0.15 ± 0.13	$\mathbf{0.42 \pm 0.09}$	$\mathbf{0.67 \pm 0.05}$	$\mathbf{0.79 \pm 0.02}$
	SPG1_STRSG_Wu	0.14 ± 0.25	$\mathbf{0.31 \pm 0.21}$	$\mathbf{0.34 \pm 0.11}$	$\mathbf{0.46 \pm 0.13}$
	DLG4_HUMAN_Faure	$\mathbf{0.32 \pm 0.13}$	$\mathbf{0.39 \pm 0.07}$	$\mathbf{0.50 \pm 0.09}$	$\mathbf{0.61 \pm 0.05}$
	GRB2_HUMAN_Faure	$\mathbf{0.33 \pm 0.22}$	$\mathbf{0.49 \pm 0.06}$	$\mathbf{0.63 \pm 0.04}$	$\mathbf{0.67 \pm 0.03}$
	F7YBW8_MESOW_Ding	$\mathbf{0.54 \pm 0.23}$	$\mathbf{0.57 \pm 0.16}$	0.61 ± 0.17	0.63 ± 0.20
ESM layer	GFP_AEQVI_Sarkisyan	0.05 ± 0.09	0.11 ± 0.11	0.23 ± 0.06	0.29 ± 0.07
	SPG1_STRSG_Olson	$\mathbf{0.20 \pm 0.19}$	0.34 ± 0.07	0.63 ± 0.04	0.78 ± 0.02
	SPG1_STRSG_Wu	$\mathbf{0.17 \pm 0.32}$	0.30 ± 0.23	0.30 ± 0.09	0.33 ± 0.22
	DLG4_HUMAN_Faure	0.22 ± 0.14	0.33 ± 0.08	0.47 ± 0.07	0.55 ± 0.11
	GRB2_HUMAN_Faure	$\mathbf{0.33 \pm 0.22}$	0.44 ± 0.08	0.61 ± 0.04	$\mathbf{0.67 \pm 0.03}$
	F7YBW8_MESOW_Ding	0.51 ± 0.23	0.54 ± 0.21	0.59 ± 0.17	$\mathbf{0.64 \pm 0.20}$
ESM logits	GFP_AEQVI_Sarkisyan	$\mathbf{0.13 \pm 0.10}$	0.13 ± 0.13	0.22 ± 0.06	0.30 ± 0.03
	SPG1_STRSG_Olson	0.16 ± 0.12	0.29 ± 0.09	0.42 ± 0.06	0.58 ± 0.03
	SPG1_STRSG_Wu	0.10 ± 0.13	0.07 ± 0.24	0.08 ± 0.23	0.26 ± 0.28
	DLG4_HUMAN_Faure	0.13 ± 0.11	0.15 ± 0.08	0.22 ± 0.12	0.34 ± 0.10
	GRB2_HUMAN_Faure	0.15 ± 0.17	0.30 ± 0.07	0.50 ± 0.08	0.59 ± 0.03
	F7YBW8_MESOW_Ding	0.37 ± 0.41	0.52 ± 0.32	$\mathbf{0.64 \pm 0.19}$	$\mathbf{0.64 \pm 0.20}$

Table 5: Average Spearman ρ across all low-N regimes under position extrapolation.

Method	**DMS**	N **= 8** $\uparrow$	N **= 24** $\uparrow$	N **= 96** $\uparrow$	N **= 384** $\uparrow$
SAE	GFP_AEQVI_Sarkisyan	0.10 ± 0.09	0.13 ± 0.11	$\mathbf{0.25 \pm 0.09}$	0.26 ± 0.15
	SPG1_STRSG_Olson	$\mathbf{0.18 \pm 0.15}$	0.36 ± 0.28	$\mathbf{0.54 \pm 0.10}$	$\mathbf{0.65 \pm 0.09}$
	SPG1_STRSG_Wu	0.11 ± 0.32	0.08 ± 0.26	$\mathbf{0.15 \pm 0.28}$	$\mathbf{0.25 \pm 0.23}$
	DLG4_HUMAN_Faure	$\mathbf{0.37 \pm 0.20}$	$\mathbf{0.52 \pm 0.10}$	$\mathbf{0.53 \pm 0.10}$	0.57 ± 0.08
	GRB2_HUMAN_Faure	$\mathbf{0.28 \pm 0.20}$	0.27 ± 0.24	$\mathbf{0.51 \pm 0.08}$	0.48 ± 0.11
	F7YBW8_MESOW_Ding	$\mathbf{0.09 \pm 0.46}$	0.10 ± 0.47	0.06 ± 0.50	0.29 ± 0.33
ESM layer	GFP_AEQVI_Sarkisyan	0.04 ± 0.10	0.09 ± 0.09	0.23 ± 0.06	0.25 ± 0.12
	SPG1_STRSG_Olson	0.18 ± 0.19	$\mathbf{0.42 \pm 0.32}$	0.46 ± 0.18	0.55 ± 0.15
	SPG1_STRSG_Wu	$\mathbf{0.16 \pm 0.29}$	$\mathbf{0.12 \pm 0.24}$	0.14 ± 0.40	0.20 ± 0.30
	DLG4_HUMAN_Faure	0.15 ± 0.38	0.41 ± 0.24	0.41 ± 0.23	$\mathbf{0.58 \pm 0.07}$
	GRB2_HUMAN_Faure	0.25 ± 0.21	0.24 ± 0.21	0.40 ± 0.21	0.40 ± 0.09
	F7YBW8_MESOW_Ding	0.05 ± 0.54	0.07 ± 0.51	0.25 ± 0.37	0.05 ± 0.38
ESM logits	GFP_AEQVI_Sarkisyan	$\mathbf{0.12 \pm 0.05}$	$\mathbf{0.13 \pm 0.10}$	0.21 ± 0.09	$\mathbf{0.26 \pm 0.07}$
	SPG1_STRSG_Olson	0.17 ± 0.20	0.30 ± 0.19	0.41 ± 0.16	0.50 ± 0.16
	SPG1_STRSG_Wu	0.13 ± 0.26	0.03 ± 0.19	0.14 ± 0.27	0.05 ± 0.16
	DLG4_HUMAN_Faure	0.00 ± 0.14	0.14 ± 0.15	0.26 ± 0.20	0.39 ± 0.10
	GRB2_HUMAN_Faure	0.20 ± 0.16	$\mathbf{0.27 \pm 0.14}$	0.41 ± 0.10	$\mathbf{0.49 \pm 0.08}$
	F7YBW8_MESOW_Ding	0.07 ± 0.51	$\mathbf{0.44 \pm 0.33}$	$\mathbf{0.34 \pm 0.38}$	$\mathbf{0.39 \pm 0.19}$

Table 6: Average Spearman ρ across all low-N regimes under regime extrapolation.

Method	**DMS**	N **= 8** $\uparrow$	N **= 24** $\uparrow$	N **= 96** $\uparrow$	N **= 384** $\uparrow$
SAE	GFP_AEQVI_Sarkisyan	**0.11** $\pm$ **0.10**	**0.23** $\pm$ **0.13**	**0.36** $\pm$ **0.05**	**0.56** $\pm$ **0.05**
	SPG1_STRSG_Olson	0.21 $\pm$ 0.14	**0.39** $\pm$ **0.06**	**0.69** $\pm$ **0.05**	**0.84** $\pm$ **0.01**
	SPG1_STRSG_Wu	**0.14** $\pm$ **0.16**	0.22 $\pm$ 0.11	**0.27** $\pm$ **0.10**	**0.32** $\pm$ **0.04**
	DLG4_HUMAN_Faure	**0.31** $\pm$ **0.20**	**0.39** $\pm$ **0.14**	**0.58** $\pm$ **0.09**	0.67 $\pm$ 0.06
	GRB2_HUMAN_Faure	**0.34** $\pm$ **0.11**	0.39 $\pm$ 0.15	0.65 $\pm$ 0.07	**0.77** $\pm$ **0.01**
	F7YBW8_MESOW_Ding	**0.37** $\pm$ **0.20**	0.53 $\pm$ 0.08	0.60 $\pm$ 0.06	0.65 $\pm$ 0.03
ESM layer	GFP_AEQVI_Sarkisyan	0.06 $\pm$ 0.10	0.17 $\pm$ 0.15	0.32 $\pm$ 0.06	0.49 $\pm$ 0.04
	SPG1_STRSG_Olson	**0.23** $\pm$ **0.09**	0.38 $\pm$ 0.08	0.64 $\pm$ 0.05	0.83 $\pm$ 0.01
	SPG1_STRSG_Wu	0.14 $\pm$ 0.18	**0.23** $\pm$ **0.11**	0.22 $\pm$ 0.08	0.29 $\pm$ 0.06
	DLG4_HUMAN_Faure	0.25 $\pm$ 0.14	0.34 $\pm$ 0.17	0.49 $\pm$ 0.13	**0.70** $\pm$ **0.05**
	GRB2_HUMAN_Faure	0.30 $\pm$ 0.09	**0.40** $\pm$ **0.14**	**0.65** $\pm$ **0.06**	0.76 $\pm$ 0.01
	F7YBW8_MESOW_Ding	0.35 $\pm$ 0.17	0.52 $\pm$ 0.09	0.59 $\pm$ 0.05	**0.68** $\pm$ **0.02**
ESM logits	GFP_AEQVI_Sarkisyan	0.11 $\pm$ 0.27	0.17 $\pm$ 0.15	0.24 $\pm$ 0.17	0.40 $\pm$ 0.04
	SPG1_STRSG_Olson	0.16 $\pm$ 0.13	0.25 $\pm$ 0.09	0.42 $\pm$ 0.03	0.56 $\pm$ 0.02
	SPG1_STRSG_Wu	0.04 $\pm$ 0.09	0.14 $\pm$ 0.07	0.16 $\pm$ 0.05	0.23 $\pm$ 0.03
	DLG4_HUMAN_Faure	0.16 $\pm$ 0.12	0.27 $\pm$ 0.07	0.34 $\pm$ 0.09	0.36 $\pm$ 0.07
	GRB2_HUMAN_Faure	0.05 $\pm$ 0.12	0.32 $\pm$ 0.06	0.48 $\pm$ 0.04	0.59 $\pm$ 0.03
	F7YBW8_MESOW_Ding	0.35 $\pm$ 0.14	**0.56** $\pm$ **0.08**	**0.60** $\pm$ **0.04**	0.64 $\pm$ 0.02

Table 7: Average Spearman ρ across all low-N regimes under score extrapolation.

Method	**DMS**	N **= 8** $\uparrow$	N **= 24** $\uparrow$	N **= 96** $\uparrow$	N **= 384** $\uparrow$
SAE	GFP_AEQVI_Sarkisyan	0.01 $\pm$ 0.09	0.01 $\pm$ 0.05	0.02 $\pm$ 0.04	**0.02** $\pm$ **0.03**
	SPG1_STRSG_Olson	0.04 $\pm$ 0.13	0.03 $\pm$ 0.10	0.12 $\pm$ 0.10	0.09 $\pm$ 0.04
	SPG1_STRSG_Wu	0.07 $\pm$ 0.07	**0.04** $\pm$ **0.07**	0.10 $\pm$ 0.09	0.18 $\pm$ 0.05
	DLG4_HUMAN_Faure	**0.05** $\pm$ **0.09**	0.04 $\pm$ 0.08	0.01 $\pm$ 0.06	0.06 $\pm$ 0.05
	GRB2_HUMAN_Faure	0.01 $\pm$ 0.06	**0.16** $\pm$ **0.11**	**0.20** $\pm$ **0.07**	0.27 $\pm$ 0.05
	F7YBW8_MESOW_Ding	**0.17** $\pm$ **0.24**	0.00 $\pm$ 0.31	0.22 $\pm$ 0.20	0.27 $\pm$ 0.10
ESM layer	GFP_AEQVI_Sarkisyan	**0.01** $\pm$ **0.06**	0.01 $\pm$ 0.05	0.01 $\pm$ 0.04	0.01 $\pm$ 0.04
	SPG1_STRSG_Olson	**0.06** $\pm$ **0.08**	**0.03** $\pm$ **0.08**	**0.12** $\pm$ **0.08**	**0.13** $\pm$ **0.05**
	SPG1_STRSG_Wu	**0.11** $\pm$ **0.11**	0.02 $\pm$ 0.08	**0.11** $\pm$ **0.08**	**0.23** $\pm$ **0.06**
	DLG4_HUMAN_Faure	0.01 $\pm$ 0.09	**0.04** $\pm$ **0.06**	**0.03** $\pm$ **0.07**	**0.07** $\pm$ **0.07**
	GRB2_HUMAN_Faure	**0.02** $\pm$ **0.07**	0.14 $\pm$ 0.11	0.18 $\pm$ 0.06	**0.30** $\pm$ **0.04**
	F7YBW8_MESOW_Ding	0.14 $\pm$ 0.24	0.03 $\pm$ 0.31	0.23 $\pm$ 0.19	0.37 $\pm$ 0.09
ESM logits	GFP_AEQVI_Sarkisyan	0.01 $\pm$ 0.08	**0.05** $\pm$ **0.08**	**0.06** $\pm$ **0.04**	0.00 $\pm$ 0.06
	SPG1_STRSG_Olson	0.00 $\pm$ 0.11	0.01 $\pm$ 0.09	0.09 $\pm$ 0.10	0.05 $\pm$ 0.06
	SPG1_STRSG_Wu	0.02 $\pm$ 0.06	0.03 $\pm$ 0.07	0.08 $\pm$ 0.05	0.13 $\pm$ 0.05
	DLG4_HUMAN_Faure	0.05 $\pm$ 0.10	0.01 $\pm$ 0.07	0.02 $\pm$ 0.04	0.00 $\pm$ 0.04
	GRB2_HUMAN_Faure	0.02 $\pm$ 0.08	0.04 $\pm$ 0.08	0.12 $\pm$ 0.05	0.23 $\pm$ 0.04
	F7YBW8_MESOW_Ding	0.15 $\pm$ 0.30	**0.03** $\pm$ **0.22**	**0.30** $\pm$ **0.21**	**0.39** $\pm$ **0.14**